

K2-30b

R-2541

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_230903 · created 2026-08-26T00:06:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460190.948 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.046
Transit depth (Rp/Rs)^2	1.5 +/- 1.12 [%]
Orbital Inclination (inc)	88.5 +/- 2.34
Airmass coefficient 1 (a1)	1.126 +/- 0.042
Airmass coefficient 2 (a2)	-0.105 +/- 0.039
Scatter in the residuals of the lightcurve fit is	3.75 %
Best Comparison Star	#3 - [372, 242]
Optimal Aperture	2.62
Optimal Annulus	9.26
Transit Duration (day)	0.093 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.046 vs 0.1266 (deviation 0.07σ)
Duration fit vs published	0.093 d vs 0.0971 d (deviation 0.15σ)
Mid-time O–C	3.9 min
Depth SNR	2.5
Residual scatter	3.75 %

Light curve

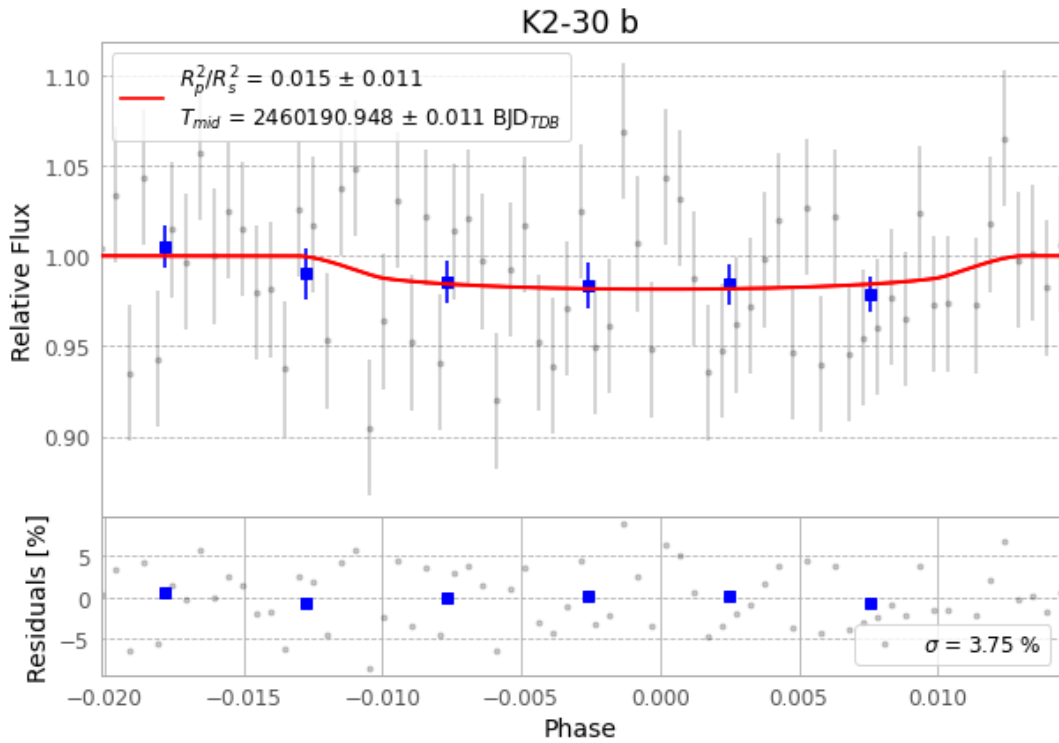


Figure 1 — FinalLightCurve_K2-30 b_2023-09-03.png (SHA-256 7f51d12ef4bbfca2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 266], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 14178f4ac568c3c7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), K2-30230903084215.FITS → K2-30230903120615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-230903.log (2caae36523c3...), FinalLightCurve_K2-30 b_2023-09-03.png (7f51d12ef4bb...), FinalLightCurve_K2-30 b_2023-09-03.csv (6a3bbd6e42b6...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:06Z.